

Final paper

Alyssa Hagele, Abby Malech, Henderson Vo

Introduction

Restoration efforts at the North Campus Open Space (NCOS) are trying to increase species richness across major habitats, where key species are driving changes in composition, habitat stability, and biodiversity. 10 native and non-native plants were selected for this study based on high observations, presence during all monitoring years (2021-2025), and frequent mention in the 2024 monitoring report and native plant catalog (CCBER). This study specifically looks at how species responded to a prescribed burn that happened in Fall 2023 as well as the effects of a year with heavier rainfall in 2024. Therefore, the main questions for this study include, **“how has percent cover of key native species and overall native species richness changed through time in grasslands compared to non-native species?”** and **“What impact does the 2023 fire and 2024 water year have on species richness and cover for both native and non-native grassland species?”**

Our native species of interest are: *Deinandra fasciculata* *Distichlis spicata* *Erigeron canadensis*
Stipa pulchra
Symphyotrichum subulatum,

And our non-native species of interest are: *Bromus hordeaceus* *Festuca myuros*
Festuca perennis
Medicago polymorpha *Polypogon monspeliensis*

This study will pay close attention to *Stipa pulchra* — California’s state grass — as well as *Festuca perennis*, a non-native grass, as these species are greatly abundant across all eight grassland transects. Lastly, *Deinandra fasciculata* is a species endemic to the Santa Barbara region and was stated to have “responded well to the fire”, which might indicate that native species perform better after a prescribed burn (Rickard 2024). The monitoring report also stated that *S. pulchra* was predicted to respond well to the fires. Other factors besides native/non-native status will be taken into account in this study, such as perennial versus annual grasses. Monitoring plant cover and richness over time can best inform restoration experts how certain intentional or unintentional events affect overall restoration progress. A study conducted on a grassland habitat found that species show positive responses to water availability extremes as

net primary productivity (NPP) increased during wet years. Additionally, functional diversity was considered an indicator of healthier ecosystems and more stable NPP, which can be informed by *structural* diversity, such as total cover and richness (Alexander et al., 2023). This is significant in the case of NCOS, as increasing native species abundance and diversity can enhance ecosystem health and resilience.

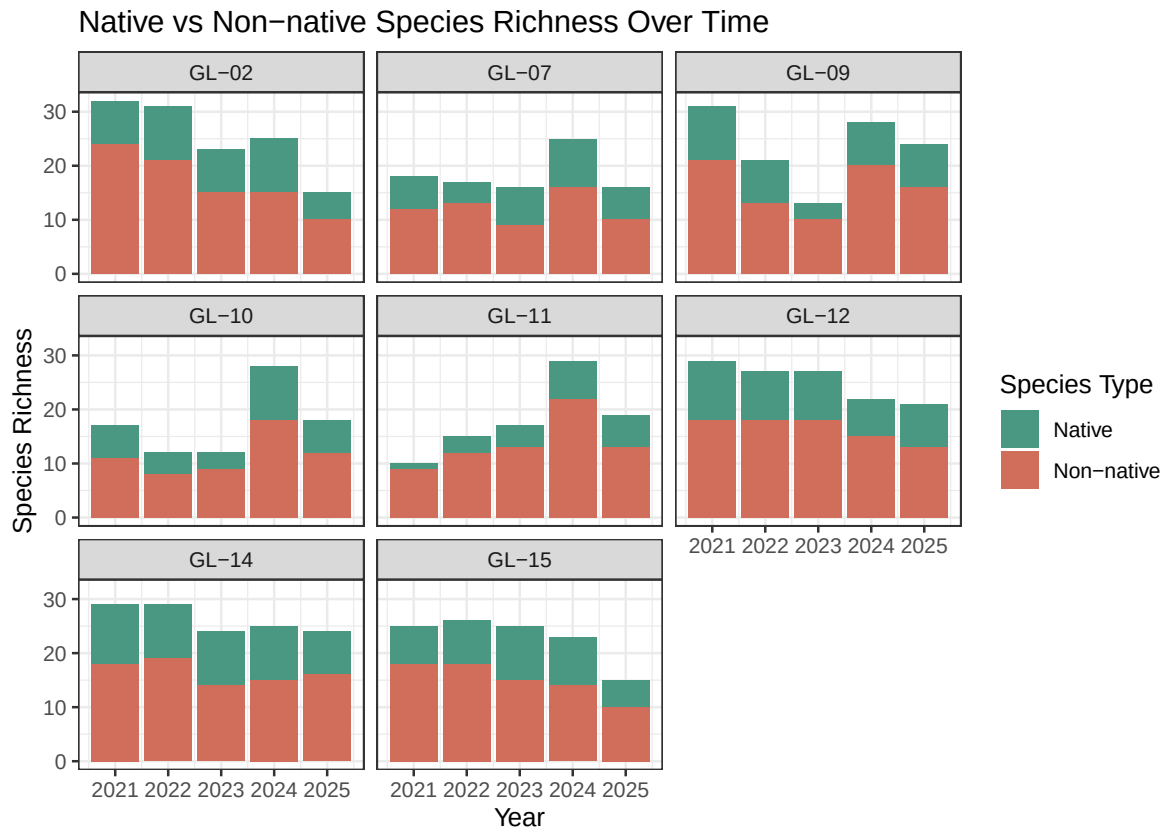
Methods

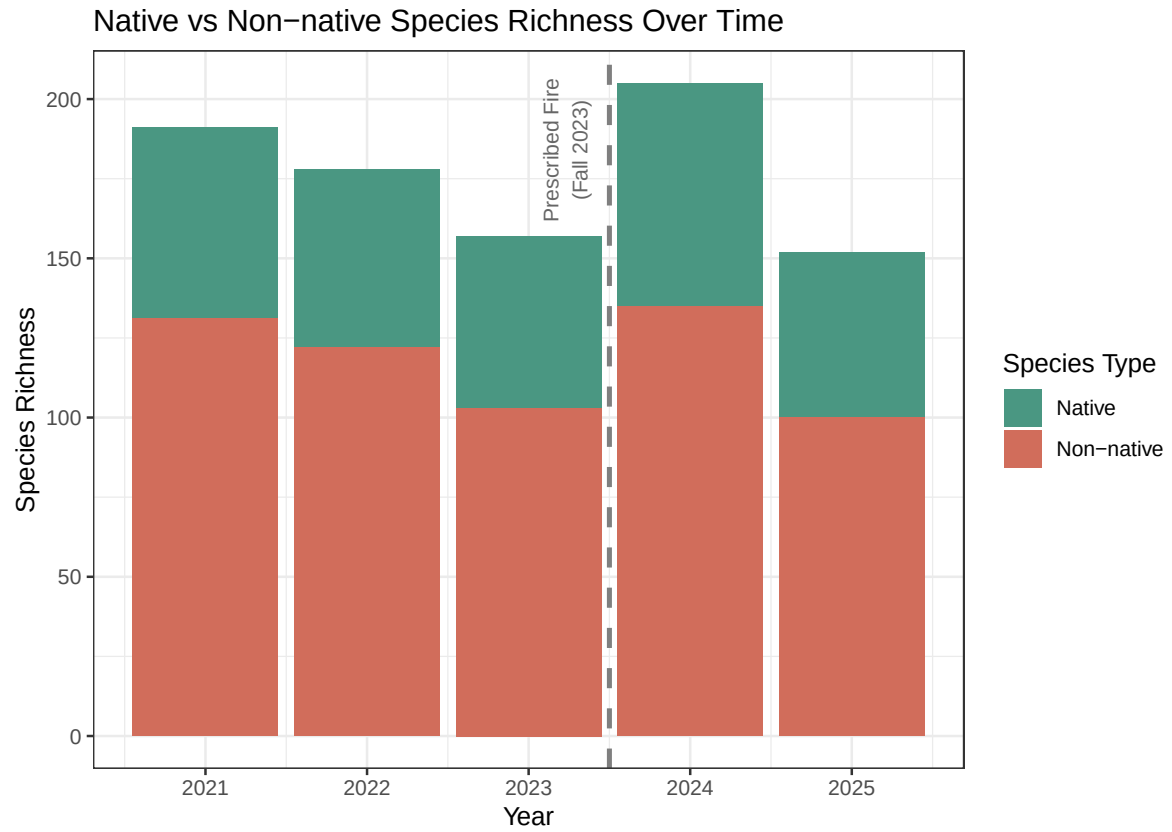
To answer our research question, we used a vegetation dataset sourced from the Cheadle Center for Biodiversity and Ecological Restoration (CCBER), and provided to us by the ENV5 193DD course. This dataset spans 5 years (2021-2025) of vegetation data from across several habitats in North Campus Open Space (NCOS) including grasslands, mosaic habitats, wetlands, open water, woodlands, and shrublands.

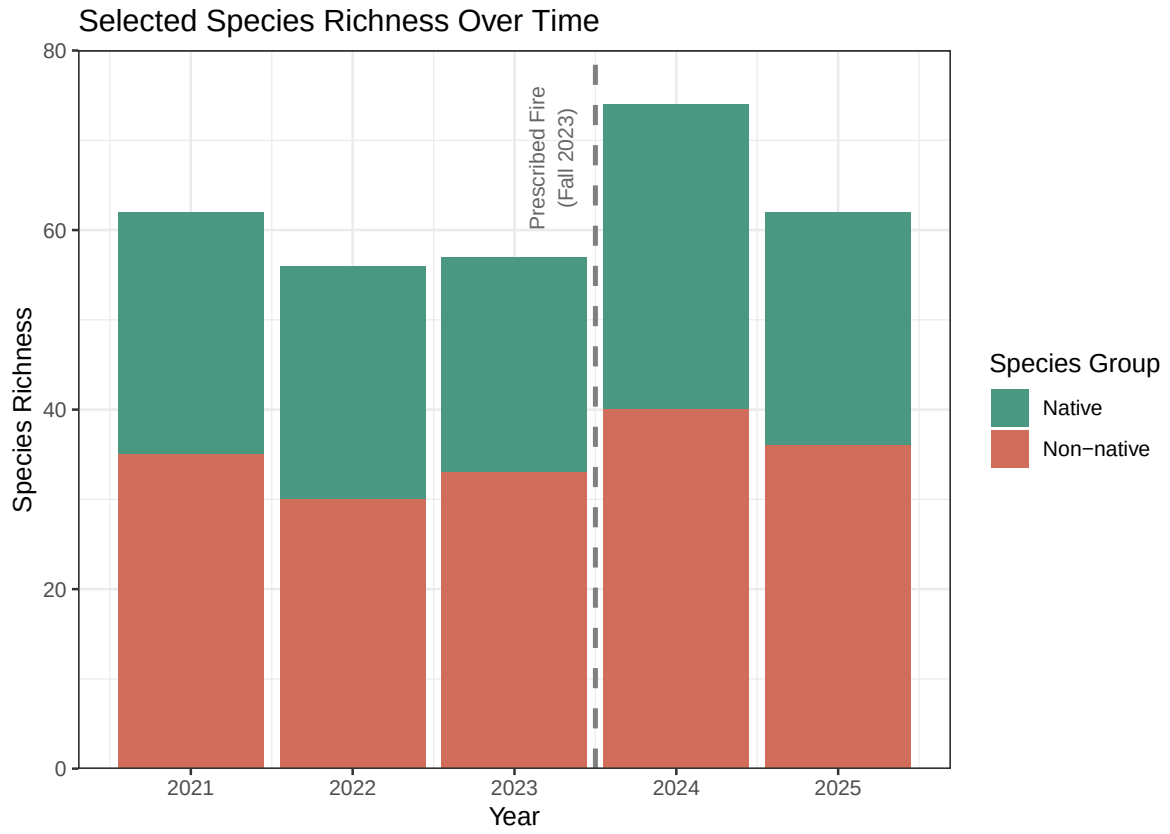
We chose to focus on native perennial grasslands, which encompasses measurements from 8 different transects across the 16.8-acre mesa. The method of data collection used for vegetation surveys is the Quadrat Transects (QT) method. This method involves the use of 11 one-square-meter quadrats placed by a 30-meter transect, with quadrats alternating between the left and right side of the transect line every 3 meters. “The quadrats are subdivided into 100 ten-centimeter squares and the percent cover is estimated [by human monitors] for each species in the quadrat,” (Rickard 2024).

We will not be performing a formal statistical analysis for our project because we are primarily exploring a correlative relationship rather than predicting trends for species cover and richness. However, we will be identifying major events that might influence a directional response in vegetation percent cover and species richness, and examining whether a response was captured in our dataset. Major events include the prescribed burn in Fall 2023 because of its effects as an intermediate disturbance event, as well as wet (2023, 2024) and dry (2021, 2022) water years because water is a limiting resource for key native grass species (*S. pulchra*). These factors are potential influences on how percent cover and species richness change over time.

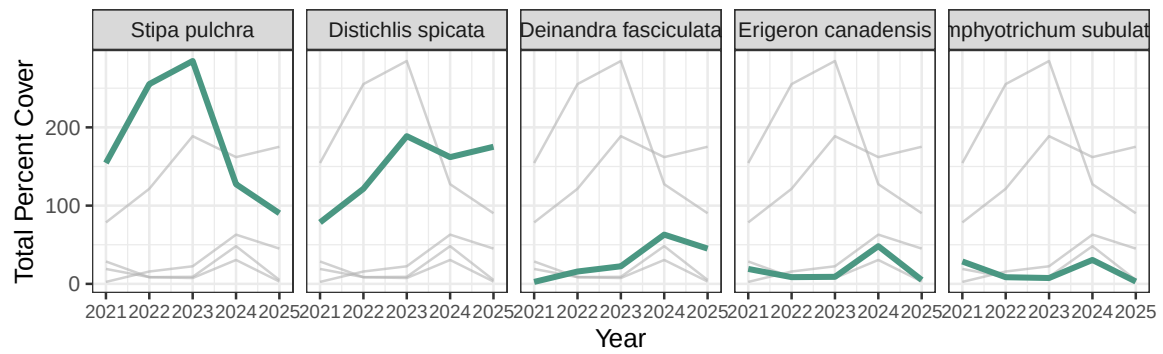
Results



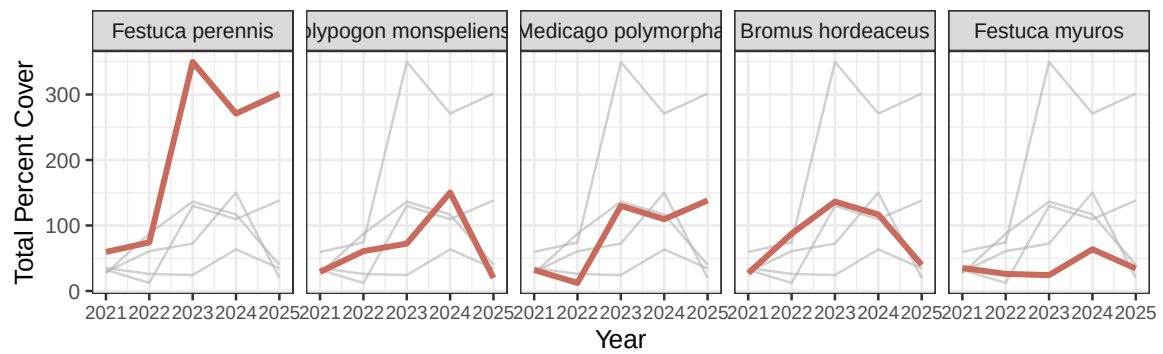




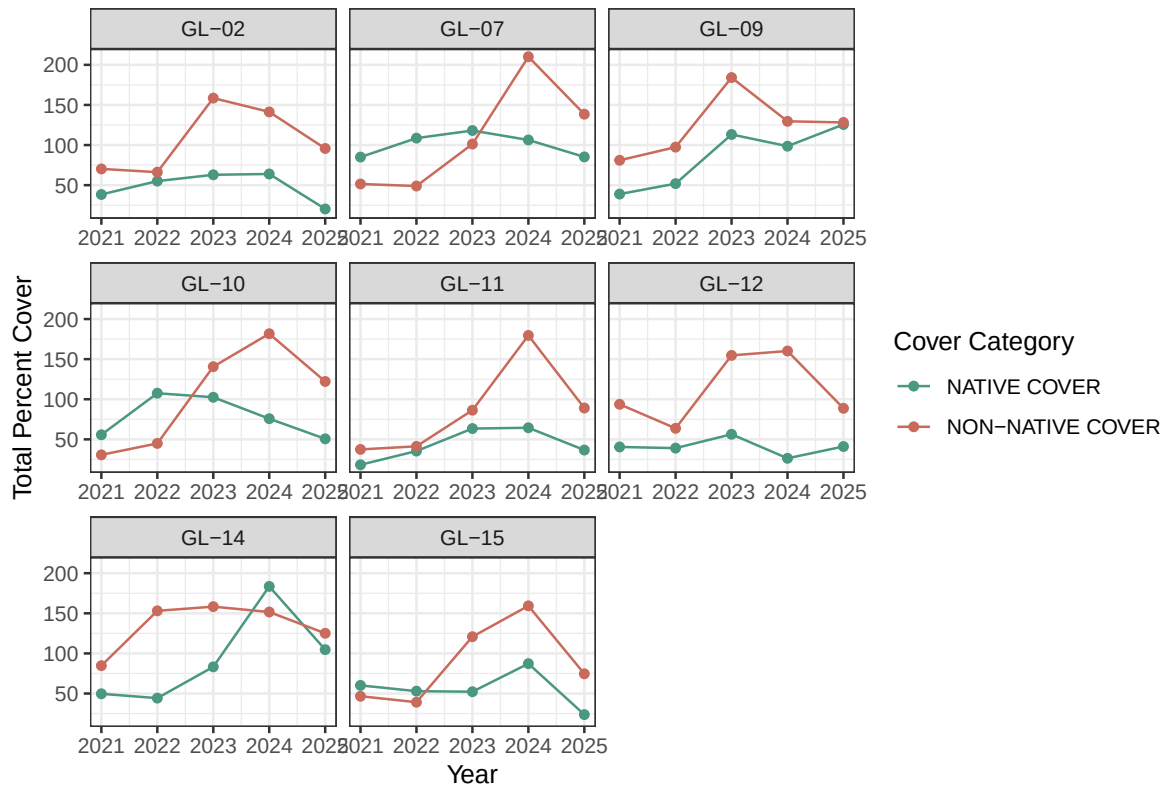
Selected Native Species Percent Cover Over Time



Selected Non-native Species Percent Cover Over Time

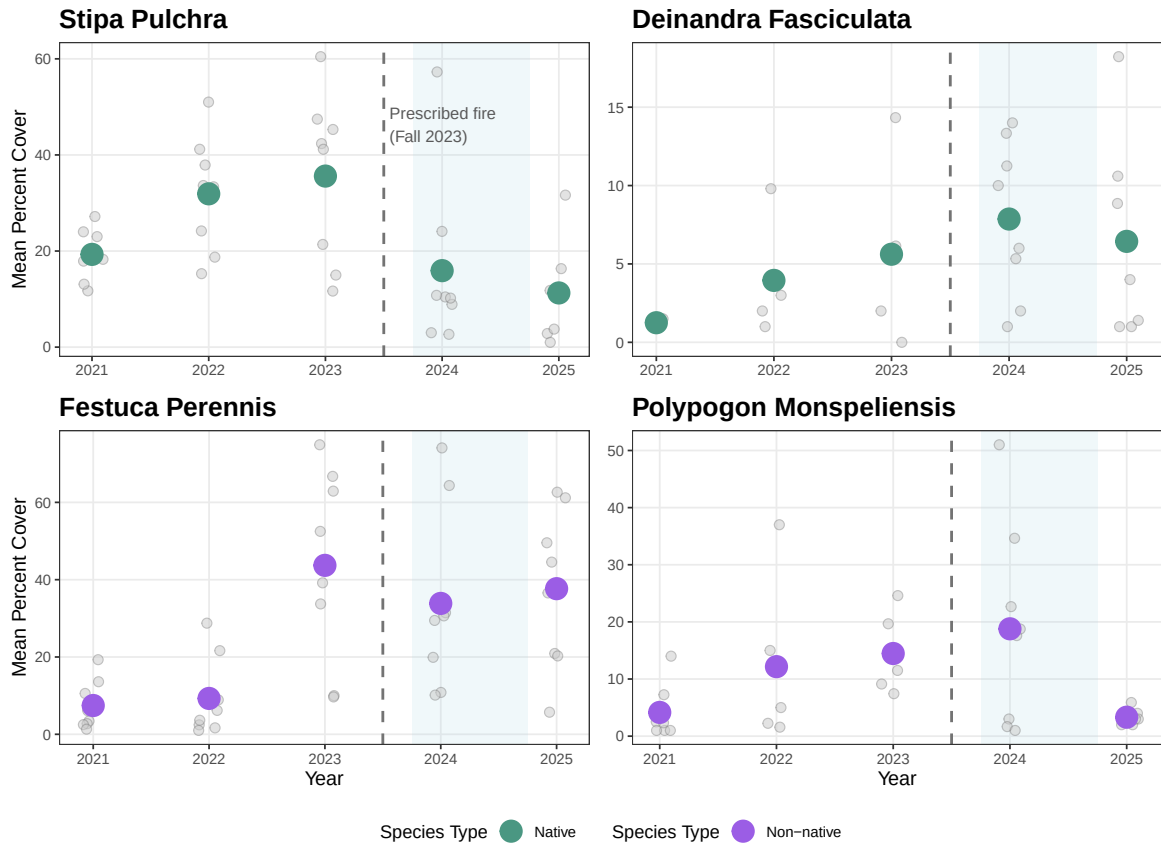


Native vs Non-Native Percent Cover Over Time Across GL Transects



Percent Cover of Four Keys Species Through Time

Blue shading indicates the 2024 water year (143% of normal precipitation).



Discussion

References

Rickard, A. 2024. "North Campus Open Space Restoration Project Monitoring Report: Year 7, December 2024." *UC Santa Barbara: Cheadle Center for Biodiversity and Ecological Restoration*. <https://escholarship.org/uc/item/16p046mb>.